CSD 204

Project Statement

Total Marks: 100

In this project, you can develop an efficient **parallel scheduling algorithm** for distributing tasks across multiple processors in a multi-core system. One can also think of some novel **process synchronization** algorithms. You can also explore and propose efficient **memory management** techniques using *machine learning* or so. You can also pick to develop efficient and novel **file system** algorithms. Then, you also have to demonstrate how this algorithm is better than the algorithms covered in the literature w.r.t some metrics such as performance, throughput, latency, etc. You can demonstrate the efficiency of the newly implemented algorithm through experiments. *You are also free to choose your own project topic related to OS and discuss it with me before starting it*.

This is a group project with a maximum of **four members** per group.

Submission Documents: As a part of this project, each group would have to submit a report consisting of the various results obtained.

1. Explain the related work in the report. They must explain the progress achieved in the area of the problem that they have chosen.

2. In addition to the report, groups would also have to submit their source code, which should be commented on and indented.

Tentative Project Schedule: The following is the schedule for the project.

1. **22nd Feb 2026:** Information regarding your group and the title of your project. Along with a maximum 250 words project statement, including what you will do in the project. A corresponding Google form has been created and shared: https://forms.gle/Jp51ZnEyS7ECQWK57 **[5 Marks]**

2. **15th March** Preliminary report of your project in 2 pages. You should do the through literature survey. You can discuss the report with us. **[20 marks]**

3. **30th March**: Discuss the **progress of the project, write the pseudocode** of the proposed algorithm **& preliminary results with me & Dr. Nazil in detail during the respective lab (31st March to 4th April)**. We will share the schedule for the project progress discussion of the respective group. You have to stick to that allocated schedule; Otherwise, we will not evaluate **your project** & **preliminary results**. **[20 Marks]**

4. **19th April 2025**: Submission of the final code and report **(IEEE format) [40 Marks]**

5. **20th to 24th April 2025:** Final project evaluations & viva. We will share the schedule for the final project discussion of the respective group. You have to stick to that allocated schedule; Otherwise, we will not evaluate **your project**. **[15 Marks]**

Note: We will have a continuous discussion of the project throughout the semester.

Plagiarism Policy: Plagiarism will strictly not be tolerated. In case, the code submitted by a group is found to have been copied from another group, then both the group submissions will not be considered. No exceptions will be made.